

Day 104 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in History & National Movement (9): Lothal's First Major Excavation in 65 Years

READING MATERIAL

Covers the renewed excavation at Lothal — the ancient Harappan port city famous for the world's oldest known dockyard — its first major dig in 65 years.

Q: When did active excavation resume at Lothal, and why is this notable?

A: Active excavation began in March 2025 (orders issued January 2025) — the first MAJOR dig at the site in 65 years; the previous large-scale excavation was conducted between 1955-1962.

Q: What factors drove this renewed excavation effort?

A: Modern technological advances enabling more precise analysis; previously unexplored areas from the original dig; the development of the National Maritime Heritage Complex project at the site; and a goal of creating 'live excavation alongside museum exhibitions' for visitors.

Q: What are some key discoveries from this 2025-26 excavation?

A: Tiny gold particles (suggesting metalworking/trade activity); 3 seals, including one depicting a UNICORN; bone fragments and fish remains (otoliths); drill bits used for bead-making; brick structures including ramp flooring; and terracotta artifacts.

Q: What modern methods are being used in this excavation?

A: Drone surveys, digital mapping, and carbon dating of approximately 30 samples — the site has been divided into 3 research zones (A, B, C), with over 50 local workers employed.

Q: What is Lothal famous for, in standard Indus Valley Civilization history?

A: It is home to the world's oldest known dockyard — evidence of the Harappan civilization's advanced maritime trade infrastructure.

MOCK TEST (WITH ANSWERS)

1. Active excavation resumed at Lothal in which month/year, marking the first major dig in 65 years?

A. January 2024

B. March 2025 ✓

C. June 2025

D. January 2026

Explanation: March 2025. [medium, web-research]

2. The previous large-scale excavation at Lothal was conducted between:

A. 1945-1950

B. 1955-1962 ✓

C. 1965-1970

D. 1975-1980

Explanation: 1955-1962. [hard, web-research]

3. Among the 2025-26 Lothal discoveries, one of the 3 seals found depicts which animal?

A. A bull

B. A unicorn ✓

C. An elephant

D. A tiger

Explanation: A unicorn. [medium, web-research]

4. Lothal is historically famous for being home to:

A. India's oldest temple

B. The world's oldest known dockyard ✓

C. The first Harappan city discovered

D. The largest Harappan granary

Explanation: The world's oldest known dockyard. [easy, web-research]

5. What major heritage project is connected to Lothal's renewed excavation?

A. The National Maritime Heritage Complex ✓

B. The Statue of Unity project

C. The Char Dham Pariyojana

D. UNESCO World Heritage nomination

Explanation: The National Maritime Heritage Complex. [medium, web-research]

Part B – Aptitude & Mental Ability: Area

READING MATERIAL

Area is 1 of 10 named Aptitude sub-skills. Real exam evidence shows a strong preference for applied/real-world framing (fencing cost, carpet cost, crossroads, a grazing horse) over bare formula plug-ins — and frequently combines area with a real-world cost-per-unit calculation.

Q: Method — area of a triangle from coordinates: Find the area of the triangle with vertices (0,0), (2,0), (0,2).

A: When two sides lie along the axes (a right triangle at the origin), area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 2 \times 2 = 2$ sq. units. (The general coordinate formula $\frac{1}{2}|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$ also works, but recognizing the right-angle-at-origin shortcut is faster.)

Q: Real exam: A rectangular ground 80m $\times$ 60m has two equal-width cross roads (one parallel to each side). Their combined area is 675 sq.m. Find the road width.

A: If width= w , the two roads' combined area = $80w + 60w - w^2$ (subtracting the overlapping square where they cross, counted twice otherwise) = $140w - w^2$. Set equal to 675: $w^2 - 140w + 675 = 0$. Solving gives $w = 5$ (the other root, 135, is too large to be physically valid for an 80 $\times$ 60 ground).

Q: Real exam: Find the perimeter of a semicircle of radius 28cm.

A: Perimeter of a semicircle = $\pi r + 2r$ (the curved half plus the straight diameter edge). Using $\pi \approx 22/7$: $(22/7 \times 28) + (2 \times 28) = 88 + 56 = 144$ cm. (Common mistake: forgetting to add the straight diameter edge, which would wrongly give just 88cm.)

Q: Real exam: A horse is tethered by a 14m rope to one corner of a 60m $\times$ 42m rectangular field. How much area is left UNGRAZED?

A: The horse can graze a quarter-circle of radius 14m (since it's tied at a corner, with two field-edges as boundaries): grazed area = $\frac{1}{4} \times \pi \times 14^2 = \frac{1}{4} \times (22/7) \times 196 = 154$ sq.m. Total field area = $60 \times 42 = 2520$ sq.m. Ungrazed = $2520 - 154 = 2366$ sq.m.

Q: Real exam: Area of a rhombus with diagonals 6cm and 8cm?

A: Area of rhombus = $\frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 6 \times 8 = 24$ cm².

Q: Real exam: If fencing a circular park at ₹5/metre costs ₹1,100 total, find the park's radius.

A: Total fencing length (= circumference) = $₹1,100 \div ₹5/\text{m} = 220$ m. Circumference = $2\pi r \rightarrow 220 = 2 \times (22/7) \times r \rightarrow r = 220 \times 7/44 = 35$ m.

Q: Real exam: Find the cost of carpeting a room 13m long, 9m wide, using a carpet 75cm wide, at Rs.8 per metre (of carpet length).

A: Room area = $13 \times 9 = 117$ m². Carpet width = 0.75m, so length of carpet needed = Room area $\div$ carpet width = $117/0.75 = 156$ m. Cost = $156 \times \text{Rs.}8 = \text{Rs.}1,248$.

Q: Real exam (related, surface area of a 3D shape): The curved surface area of a right circular cylinder of height 7cm is 66 cm². Find its diameter.

A: CSA = $2\pi rh \rightarrow 66 = 2 \times (22/7) \times r \times 7 = 44r \rightarrow r = 1.5$ cm $\rightarrow$ diameter = 3cm.

MOCK TEST (WITH ANSWERS)

1. Find the area of the triangle with vertices (0,0), (5,0), (0,4).

A. 10 sq. units ✓

B. 20 sq. units

C. 9 sq. units

D. 18 sq. units

Explanation: Right triangle at origin: area = $\frac{1}{2} \times 5 \times 4 = 10$ sq. units. [easy, authored]

2. Find the perimeter of a semicircle of radius 21cm (use $\pi \approx 22/7$).

A. 66 cm

B. 108 cm ✓

C. 132 cm

D. 96 cm

Explanation: Perimeter = $\pi r + 2r = (22/7 \times 21) + (2 \times 21) = 66 + 42 = 108$ cm. [medium, authored]

3. Find the area of a rhombus with diagonals 10cm and 12cm.

A. 60 cm² ✓

B. 120 cm²

C. 30 cm²

D. 100 cm^2

Explanation: Area = $\frac{1}{2} \times 10 \times 12 = 60\text{ cm}^2$. [easy, authored]

4. Fencing a circular garden at ₹4 per metre costs ₹880 in total. Find the garden's radius (use $\pi \approx 22/7$).

A. 28 m

B. 35 m ✓

C. 22 m

D. 14 m

Explanation: Circumference = $880/4 = 220\text{m}$. $r = 220 \times 7 / (2 \times 22) = 1540/44 = 35\text{m}$. [medium, authored]

5. Find the cost of carpeting a room 10m long and 8m wide with a carpet 80cm wide, at Rs.10 per metre.

A. Rs.800

B. Rs.1,000 ✓

C. Rs.1,200

D. Rs.960

Explanation: Room area = 80 m^2 . Carpet length needed = $80/0.8 = 100\text{m}$. Cost = $100 \times \text{Rs.}10 = \text{Rs.}1,000$. [medium, authored]

6. A horse is tethered by a 7m rope to a corner of a $20\text{m} \times 15\text{m}$ rectangular field. Find the grazed area (use $\pi \approx 22/7$).

A. 38.5 m^2 ✓

B. 44 m^2

C. 49 m^2

D. 77 m^2

Explanation: Quarter circle of radius 7m: $\frac{1}{4} \times (22/7) \times 49 = 38.5\text{ m}^2$. [medium, authored]

Part C – General English: Poem: "The Comet" — Norman Littleford

READING MATERIAL

A science-flavoured poem describing a comet's speed, size, and the science of its glowing tail. Real coaching-site evidence tests figure-of-speech identification heavily, plus the underlying science content (ice melting into vapour forming the tail).

Q: Who wrote "The Comet", and what is the poem broadly about?

A: Norman Littleford. The poem describes a comet's speed and size, the science of how its tail forms (ice melting into vapour as it nears the sun), the potential danger if a comet came too close to Earth (atmospheric shockwaves, earthquakes), and even touches the scientific idea that cometary dust chemicals may have contributed to the origins of life.

Q: What is the poem's structure (stanzas, lines, rhyme scheme)?

A: Six stanzas of four lines each, rhyme scheme ABCB.

Q: Identify the figure of speech: "Rampaging through the heavens."

A: Cited as BOTH Personification (the comet given the active, almost violent human/animal action of 'rampaging') and Imagery (a vivid visual impression) in different sourced questions — both labels are defensible for the same line, so be ready for either framing.

Q: Identify the figure of speech: "Faster than a cheetah."

A: Simile — direct comparison using 'than' (the comparative form of 'like/as').

Q: Identify the figure of speech: "A comet in full flight."

A: Metaphor — describes the comet's motion directly as 'flight' without 'like/as'.

Q: Identify the figure of speech: "With a tail that's miles long."

A: Hyperbole — an exaggerated claim about the tail's length, for dramatic emphasis.

Q: Which line is the clearest example of hyperbole, among 'Bigger than a mountain', 'So powerful and strong', 'Causing vapor from the force', and 'The outer ice is melting'?

A: "Bigger than a mountain" — an exaggerated size comparison, the others are more literal/descriptive statements.

MOCK TEST (WITH ANSWERS)

1. "The Comet" is structured as how many stanzas, and what rhyme scheme?

A. 4 stanzas, AABB

B. 6 stanzas, ABCB ✓

C. 8 stanzas, ABAB

D. 6 stanzas, AABB

Explanation: Six stanzas of four lines, ABCB rhyme scheme. [hard, web-research]

2. "Faster than a cheetah" uses which figure of speech?

A. Metaphor

B. Simile ✓

C. Personification

D. Hyperbole

Explanation: Simile — comparison using 'than'. [easy, web-research]

3. "A comet in full flight" is an example of:

A. Simile

B. Metaphor ✓

C. Hyperbole

D. Personification

Explanation: Metaphor — motion described directly as 'flight'. [medium, web-research]

4. "With a tail that's miles long" uses which figure of speech?

A. Simile

B. Metaphor

C. Hyperbole ✓

D. Personification

Explanation: Hyperbole — exaggerated for emphasis. [medium, web-research]

5. According to the poem, what causes the comet's glowing tail to form?

A. Reflected starlight

B. Ice melting into vapour ✓

C. Magnetic radiation

D. Friction with the solar wind

Explanation: The outer ice melts into vapour, trailing behind as the tail. [medium, web-research]

6. Which of these lines is the clearest example of hyperbole?

A. Bigger than a mountain ✓

B. So powerful and strong

C. Causing vapor from the force

D. The outer ice is melting

Explanation: "Bigger than a mountain" — an exaggerated size claim. [medium, web-research]